When Labels Override Evidence

A label shortcut in an encoder decision model: its circuit, its training origin, and how to remove it

Vignesh Kanike · 2026-10-12 · model studied: convaiinnovations/laya (laya 0.4.1) · all numbers rebuilt from `release/results`

Abstract. Laya scores each answer option at its own [MASK] token in a single ModernBERT encoder pass. We show that its yes/no answers follow the *meaning of the option label words* rather than the passage: on a frozen, human-written benchmark of 600 MNLI and QNLI items, swapping "yes" and "no" flips 96.7% of the answers laya otherwise gets right (n = 516), while swapping neutral symbols flips 4.1%. Mechanistically, the [MASK] reads its label word directly through local attention, redundantly across layers, while evidence from the passage reaches the [MASK] only through the question tokens. Blocking the single attention edge class [MASK]→label cuts label-following to 2.6% at an accuracy cost of 1.8 points, while blocking the same number of random edges changes nothing; the result replicates on two sibling checkpoints and a second frozen benchmark. In a dose study of 27 retrained ModernBERT models, the shortcut appears whenever the label word identifies the option during training (planted mean 88.9% vs baseline 14.7%; exact permutation p = 0.0022), even when it is redundant with the option text, and that randomising label words in training prevents it (0.8% label-following). In RoBERTa-large the shortcut appears in planted and in most baseline seeds, so the pre-registered planted-vs-baseline difference was not significant there, while the same attention block still removes it. Three other pre-registered predictions failed: a redundant label was predicted to go unused, the guard does not transfer cleanly to every retrained ModernBERT seed, and projecting out a learned label direction removes the answer along with the label. Order sensitivity, confidence and jailbreak detection are analysed as supporting findings. Hypotheses were frozen before confirmatory runs, and claims that failed are reported as such.

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1 · Introduction

Encoder-only decision models such as laya answer a question by scoring several written options in one pass. Each yes/no option carries a label word ("no" / "yes") and a description ("the statement does not hold" / "holds"). A model that answers from the passage should not care which word labels which option. Laya does: its answer tracks the label word. The model card notes this failure; this paper explains it, shows what training produces it, and tests fixes.

Contributions.

1. **Behaviour, confirmed.** On a benchmark frozen and hashed before evaluation, laya follows swapped yes/no labels on 96.7% of correctly answered items (95% CI 95.2%–98.1%), and on never-seen wordings ("right"/"wrong": 80.4%).
2. **Circuit.** Two pathways: a direct, redundant [MASK]→label read in local attention, and an evidence relay passage → question tokens → [MASK].
3. **Fix at inference.** A one-edge-class attention guard removes the shortcut in all three laya checkpoints; matched control edges do not.
4. **Origin.** Retraining shows which data create the shortcut, a dose response, a training-time fix, and that RoBERTa-large learns the shortcut even more readily.
5. **Limits.** Four pre-registered predictions failed (redundant labels unused, guard transfer, label-direction ablation, RoBERTa planted vs baseline); we report where and why.

Secondary claims (C1 order sensitivity, C2 confidence, C4 jailbreak detection, C5 training objective) are in Section 5.

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2 · Model, data and methods

2.1 Model and instrument

Laya is ModernBERT-large (28 layers, width 1024, 16 heads; global attention every third layer from layer 0, a ±64-token window elsewhere) plus two transformer head layers and a per-option scorer. The input is `[CLS] <type> question: <instructions> [SEP]`

`[MASK] option0 [MASK] option1 … [SEP] state [SEP]`; a yes/no option renders as "`false: no, the statement does not hold`". All analysis uses an explicit re-implementation of the forward pass with named, editable sites (residual stream, attention pattern and output, MLP output, per-edge attention blocks), checked against laya's own `predict`:

Checkpoint	Max probability difference vs predict	Local window
laya	5.1e-05	±64
smolaya	5.0e-05	±64
typed	5.0e-05	±64
multilingual	4.9e-05	±64

2.2 Data

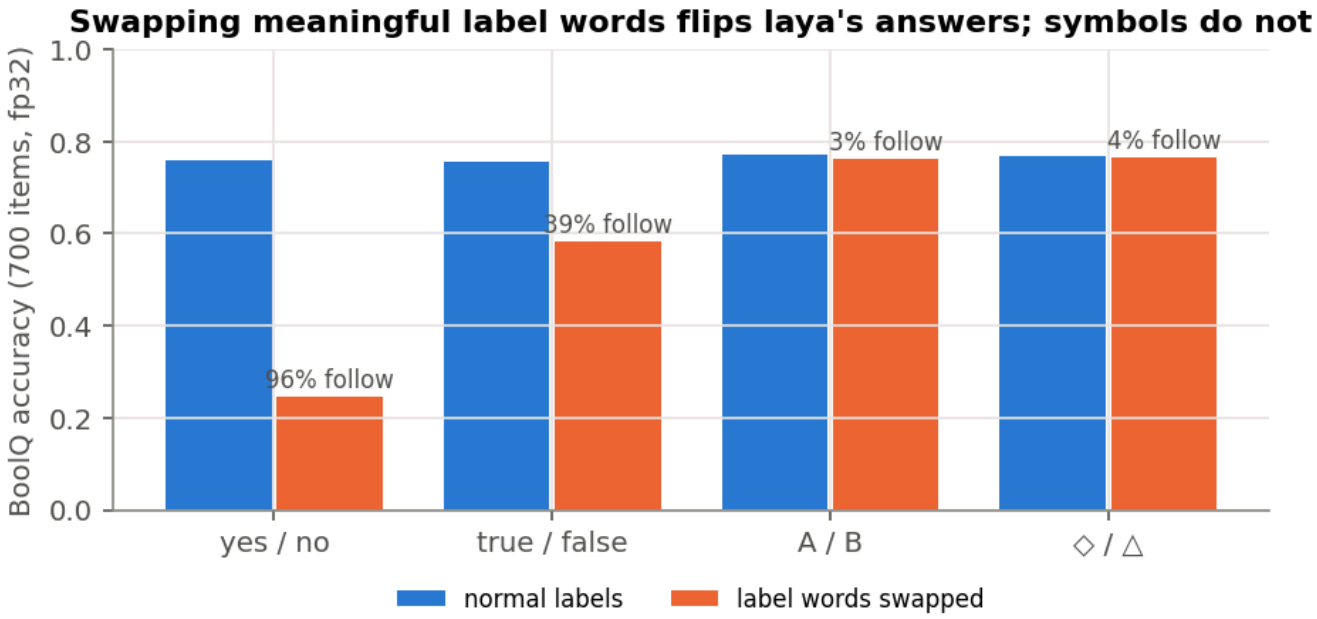
- **Exploratory:** BoolQ (700 items; 300 held out), GPT-written items.
- **Untouched human-written:** RTE (277) and MultiRC (500).
- **Benchmark 1 (confirmatory):** 600 MNLI + QNLI items never used before, frozen and hashed before evaluation (SHA-256 cea5547f…), with 519 contradicting edits, three never-used label wordings and two multi-token ones.
- **Benchmark 2 (confirmatory):** 600 further MNLI + QNLI items with no overlap (SHA-256 05c0217c…), frozen with a written prediction before any evaluation.
- Contradicting edits and some items were written by gpt-5.6-terra and checked by gpt-5.6-luna; random samples were then audited (Appendix A).

2.3 Definitions

- **Paired transitions.** Each item is scored with normal and with swapped label words; we count correct→correct, correct→wrong, wrong→correct and wrong→wrong.
- **Label-following** = n(correct→wrong) / n(correct with normal labels), with the denominator reported and a bootstrap 95% interval over items.
- **Recovery** per item = (y_patched − y_corrupt) / (y_clean − y_corrupt), y = log-odds of "holds"; items with |y_clean − y_corrupt| < 0.5 are excluded and counted; median with bootstrap interval.
- **Guard:** block attention from each yes/no [MASK] to its label tokens in the local layers. **Extended guard:** also block option-description tokens from reading the label tokens. **Controls:** the same number of random option edges, the same number of description edges, and a sham [MASK]→[CLS] edge.
- **Seed-level tests** treat each trained model as one unit: per-seed values are listed and compared with an exact two-sided permutation test.
- **Pre-registration.** Each confirmatory test's data, interventions and prediction were written to `NEXT_STEPS_PLAN.md` before it ran; from the confirmatory benchmarks on, each pre-registration is in a git commit that precedes its results. Analyses added afterwards are labelled exploratory.

3 · The label shortcut and its circuit

3.1 Behaviour



On BoolQ (fp32), swapping "yes"/"no" flips 96% of correct answers and "true"/"false" 39%, while "A"/"B" flips 3% and "◇"/"△" 4%: laya reads the meaning of the label word, not its identity.

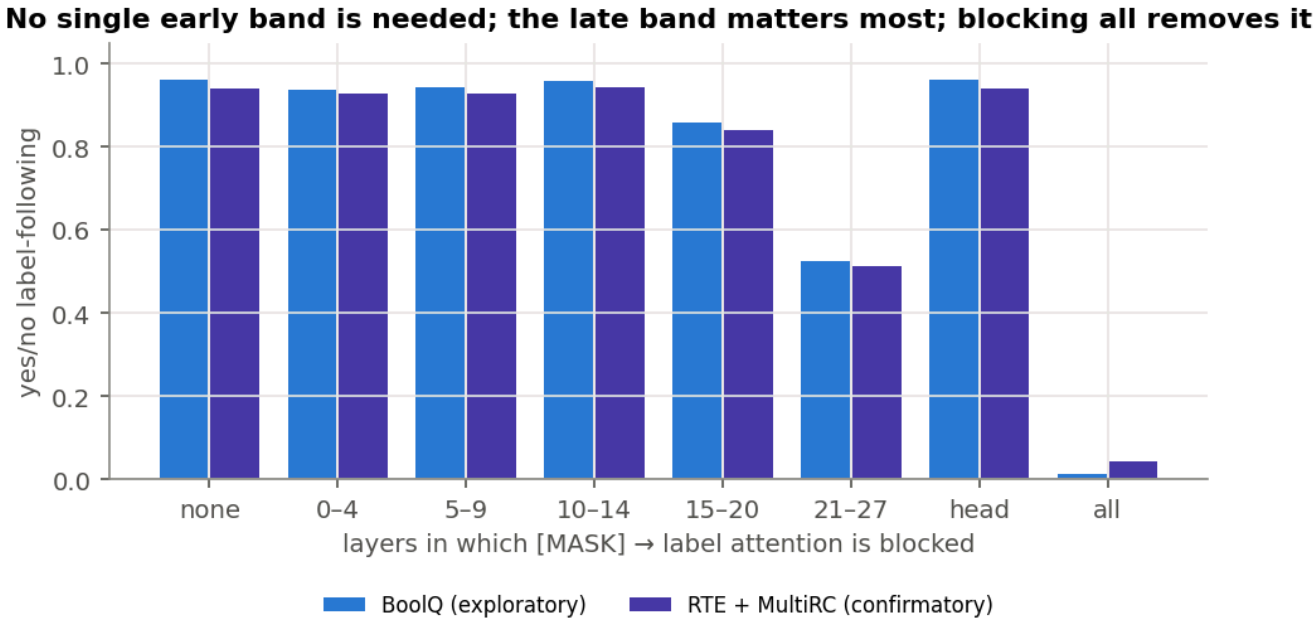
3.2 Confirmatory test and matched controls

On benchmark 1, laya's transitions are: correct→correct 17, correct→wrong 499, wrong→correct 70, wrong→wrong 14. The guard is specific: blocking the same number of other edges, or a sham edge, leaves label-following unchanged, while it costs 1.8 accuracy points and lowers jailbreak AUROC by 0.023.

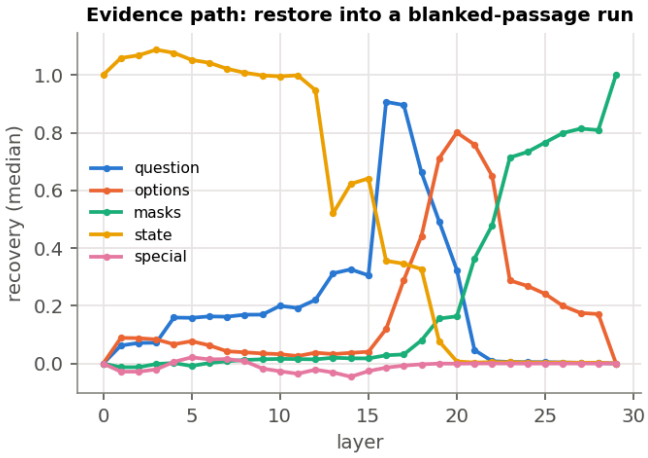
Benchmark 1, laya (fp32)	yes/no	right/wrong (new)	"yes, that is right" (new)	◇/△	Accuracy	Jailbreak / injection AUROC
None	96.7%	80.4%	96.9%	4.1%	0.860	0.940 / 0.896
Guard: [MASK]→label, local layers	2.6%	3.0%	28.8%	2.0%	0.842	0.917 / 0.872
Extended guard: + description→label	1.8%	1.6%	7.6%	2.8%	0.852	0.911 / 0.875
Control: same number of random option edges	97.1%	85.7%	98.2%	4.5%	0.858	0.940 / 0.897
Control: same number of description edges	97.9%	85.1%	98.2%	5.4%	0.855	0.939 / 0.895
Control: [MASK]→[CLS]	98.3%	91.9%	99.2%	4.1%	0.867	0.946 / 0.897
Evidence path blocked: question→passage	84.2%	72.6%	81.6%	10.6%	0.645	0.881 / 0.742

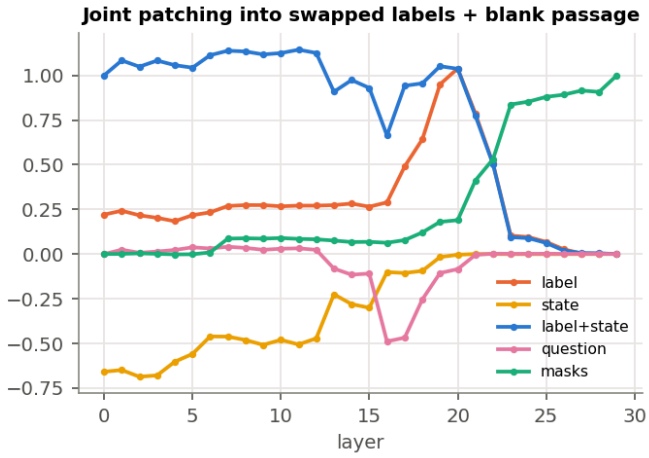
On contradicting passages the guard cuts label-following from 97.5% to 4.5%. With the passage blanked, swapping labels moves "holds" answers from 33% to 95%; with the guard, 5% to 11%.

3.3 Two pathways



Block reading of the passage by...	Accuracy, plain	Accuracy, with guard
none	0.737	0.727
masks	0.730	0.733
options	0.760	0.757
question	0.537	0.507
special	0.747	0.747
all_nonstate	0.517	0.447

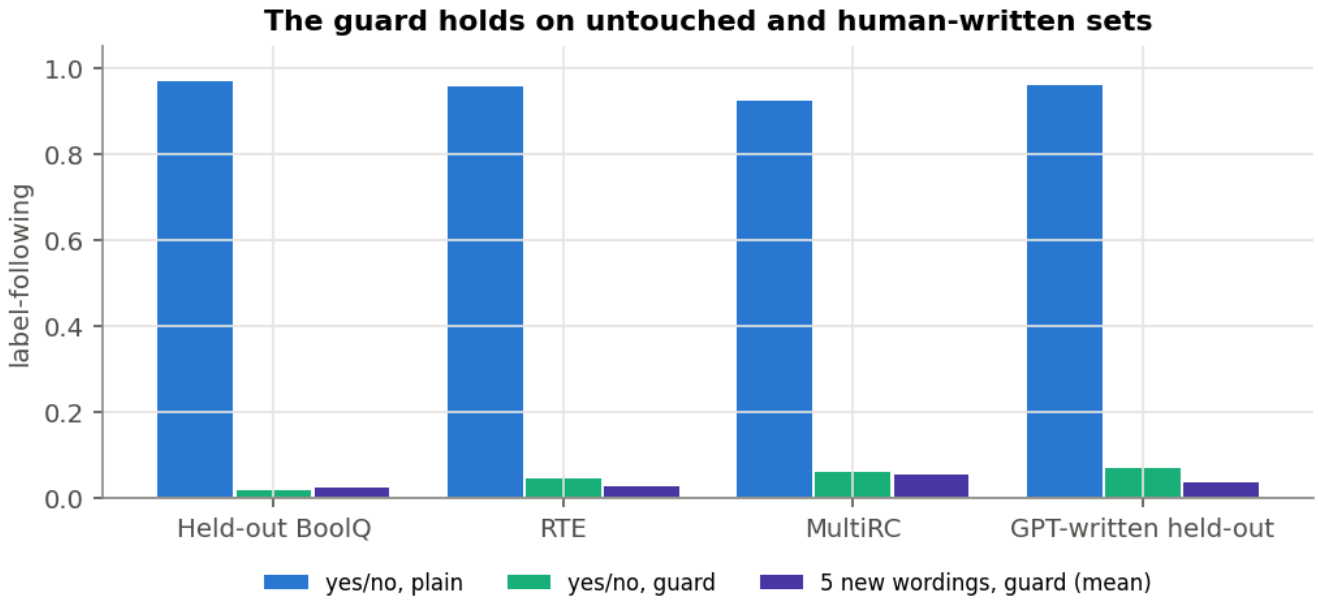




- **The label read is redundant and mostly late.** Blocking [MASK]→label in layers 10–14 alone leaves label-following at 96%; blocking layers 21–27 alone halves it (53%); blocking every local layer removes it.
- **Evidence goes through the question tokens.** Stopping them from reading the passage drops accuracy from 0.737 to 0.537; stopping the [MASK]s or option text from reading it changes nothing. On benchmark 1, restoring only the question tokens into a blanked-passage run recovers a median 0.91 [0.85, 0.96] of the decision at layer 16 (n = 275; 25 of 300 excluded for a small gap).
- **No timing race.** Label information enters the [MASK] in layers 21–23, the same layers where the evidence arrives.

3.4 The guard in deployment and across checkpoints

Installed on a stock `laya.load` agent and run through the official `predict` API (`interp/label_guard.py`), the guard leaves choice questions bit-identical (max difference 0) and moves jailbreak AUROC 0.940 → 0.919.



Untouched set (laya, fp32)	yes/no label-following, plain → guard	Accuracy, plain → guard
Held-out BoolQ (300)	97% → 2%	0.757 → 0.753
RTE (277, human-written)	96% → 4%	0.780 → 0.812
MultiRC (500, human-written)	92% → 6%	0.586 → 0.614
GPT-written held-out (292)	96% → 7%	0.767 → 0.808

Benchmark 1	yes/no, none	Guard	Extended guard, multi-token label	Random-edge control	Accuracy, none → guard
laya	96.7%	2.6%	7.6%	97.1%	0.860 → 0.842
typed	93.5%	3.9%	2.8%	95.7%	0.825 → 0.845
multilingual	88.7%	1.6%	6.5%	89.0%	0.735 → 0.743

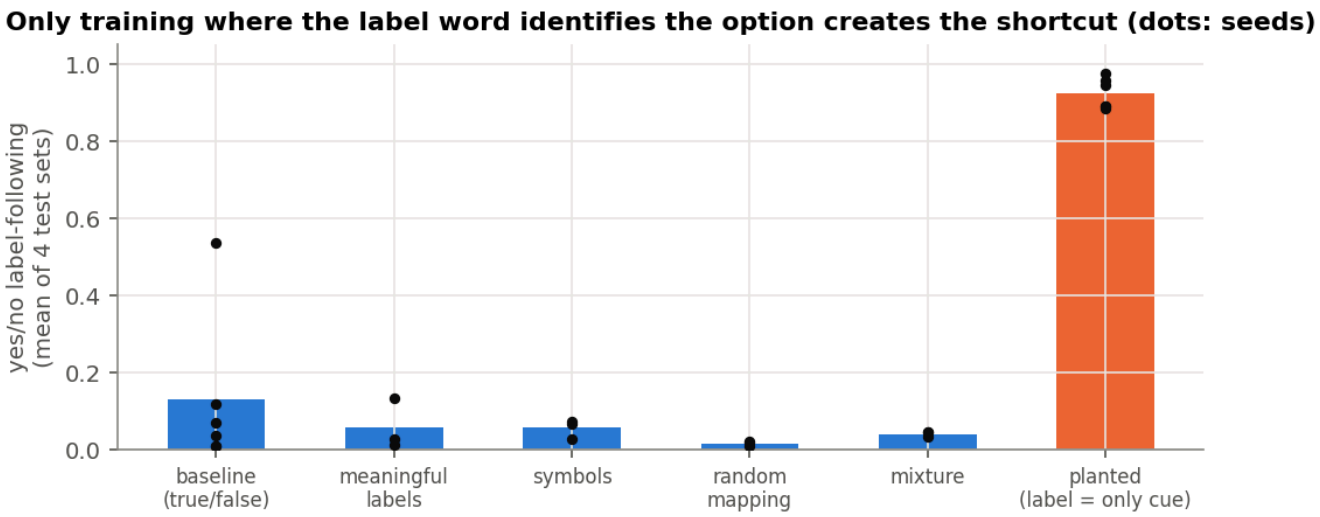
The behaviour, the guard and the two-pathway picture hold on all three checkpoints (laya, laya-typed-decisions and laya-multilingual, which has a different base model). Multi-token labels need the extended guard.

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4 · Where the shortcut comes from

4.1 Which training creates it

We trained ModernBERT-large models with laya's trainer, data (31,419 items) and loss (RLCD), changing only how the yes/no labels were presented. Laya's own training data are not public, so this shows what *can* create the shortcut, not what did.



Training regime (3 seeds; earlier test sets)	yes/no label-following	Accuracy
Baseline: default true/false	12.8% ± 20.3	0.749
Meaningful labels, aligned	5.8% ± 6.6	0.751
Meaningless symbols	5.5% ± 2.4	0.754
Meaningful labels, randomly swapped	1.4% ± 0.6	0.752
Mixture	4.0% ± 0.6	0.756
Planted: label word is the only cue	92.3% ± 4.0	0.768

Only training where the label word identifies which option means what creates the shortcut ($p = 0.000$ vs baseline); meaningful labels alone do not ($p = 0.46$).

4.2 Dose response on benchmark 1

Training arm (shuffled option order except baseline)	yes/no label-following per seed (%)	Mean (%)	Accuracy
Baseline: default labels, fixed order	0.2, 0.7, 18.1, 9.3, 54.9, 4.7	14.7	0.773
D0: label random, informative text	1.3, 0.0, 0.9	0.8	0.743
D1: label always right, informative text	2.8, 31.2, 98.5, 94.7, 92.1, 81.8	66.9	0.768
D2: label right 75%, neutral text	91.9, 41.1, 90.9	74.6	0.688
D3: label right 90%, neutral text	95.3, 92.0, 85.8	91.0	0.764
Planted: label always right, neutral text	81.2, 86.3, 84.2, 94.7, 92.3, 95.0	88.9	0.784

- **Planted vs baseline** (6 seeds each, pre-registered): every planted model exceeds every baseline model; exact $p = 0.0022$, the minimum this test allows.
- **Reliability.** A sole-cue label that is right only 75% or 90% of the time already produces strong label-following.
- **Redundancy is not protection.** When the option text also says which option is which (D1), 4 of 6 seeds still rely on the label (D1 vs baseline $p = 0.024$; vs planted $p = 0.34$). We had predicted D1 would stay low; that prediction was wrong.
- **Baselines are not immune:** one of six baseline seeds follows labels 55% of the time.

4.3 A training-time fix

Arms D0 and D1 differ only in whether the label word is reliable. When label words are assigned at random during training (D0), label-following is 1.3%, 0.0%, 0.9% across 3 seeds, against 2.8%, 31.2%, 98.5%, 94.7%, 92.1%, 81.8% for D1 (exact $p = 0.060$). Benchmark accuracy is 0.743 for D0 vs 0.768 for D1 and 0.773 for baseline. Randomising label words in training therefore prevents the shortcut at a cost of about 2 accuracy points in this setting. This was not pre-registered as a fix, and it rests on 3 seeds.

4.4 Does the guard transfer to retrained models? (pre-registered; failed)

The local-layer guard worked unevenly on retrained planted models. Before running benchmark 2 we predicted that blocking [MASK]→label in *all* layers would leave ≤ 10% label-following in the laya checkpoints and in at least 5 of 6 planted seeds, while the same number of random edges would change nothing. The laya checkpoints passed; only 3 of 6 planted seeds did, so the prediction **failed**. The control stayed within 10 points of no intervention in 9 of 9 models. Adding description→label edges (extended, all layers) reached ≤ 10% in 5 of 6 seeds.

Benchmark 2: yes/no label-following (accuracy)	None	Guard, local	Guard, all layers	Extended guard, all layers	Control: random edges, all layers
laya	96.7% (0.850)	3.8% (0.875)	4.8% (0.867)	1.1% (0.880)	96.7% (0.860)
typed	93.9% (0.820)	2.7% (0.858)	2.3% (0.855)	0.4% (0.863)	95.5% (0.817)
multilingual	89.0% (0.773)	0.9% (0.773)	0.6% (0.772)	0.2% (0.772)	89.6% (0.782)
p5_planted_s0	85.6% (0.810)	40.6% (0.788)	3.6% (0.742)	1.1% (0.772)	87.2% (0.810)
p5_planted_s1	86.6% (0.857)	77.9% (0.837)	32.7% (0.617)	23.6% (0.622)	86.9% (0.855)
p5_planted_s2	85.1% (0.797)	20.2% (0.792)	5.8% (0.770)	2.2% (0.772)	86.0% (0.798)
p5_planted_s3	94.5% (0.787)	65.5% (0.773)	10.2% (0.717)	6.7% (0.698)	94.1% (0.788)
p5_planted_s4	94.3% (0.787)	34.5% (0.777)	4.7% (0.712)	3.0% (0.728)	94.7% (0.787)
p5_planted_s5	95.0% (0.803)	73.3% (0.725)	13.7% (0.500)	1.2% (0.567)	94.8% (0.803)

Retrained models read the label through the same two edge classes as laya, in different layers and proportions per seed. In seeds 1 and 5, blocking them also costs 20–30 accuracy points: the label is entangled with normal answering.

4.5 Removing a label direction instead of an edge (exploratory)

For each layer we fitted one direction on benchmark 1: the mean residual at the "holds" [MASK] with normal labels minus with swapped labels. We then projected it out at every [MASK] in every layer, and evaluated on benchmark 2. The success criterion, fixed in advance, was label-following ≤ 10% with an accuracy drop ≤ 5 points. Direction ablation alone met it in 0 of 3 models; combined with the extended all-layer guard, in 0 of 3. Label-following does fall to zero, but only because the model stops answering: accuracy falls to 0.522, which is exactly the accuracy of always answering "does not hold" on benchmark 2 (0.522). A random direction per layer changes nothing. The direction that separates normal from swapped labels at the [MASK] is the answer direction itself: in this model, "what the label says" and "what the answer is" share one linear direction, so they cannot be separated by projection.

Benchmark 2: yes/no label-following (accuracy)	None	Direction ablation	Random-direction control	Direction ablation + extended guard
laya	96.7% (0.850)	0.0% (0.522)	96.6% (0.843)	0.0% (0.522)
p5_planted_s1	86.6% (0.857)	0.0% (0.522)	87.2% (0.845)	0.0% (0.522)
p5_planted_s5	95.0% (0.803)	0.0% (0.522)	94.4% (0.802)	0.0% (0.522)

4.6 A second architecture: RoBERTa-large

We trained option-marker models on RoBERTa-large with the same recipe (encoder learning rate 1e-5, chosen in advance after 3 of 6 runs at laya's 2.5e-5 failed to learn), 6 planted and 6 baseline seeds, all reported. 12 of 12 learned the task (BoolQ ≥ 0.70). Among them, planted label-following averages 77.6% vs 48.9% for baseline (exact p = 0.1126, 6 vs 6 seeds). The [MASK]→label block is applied in all layers (RoBERTa has no local/global split); a fidelity check confirms the unblocked path matches the plain model exactly.

The pre-registered prediction (planted > baseline, p < 0.05) **failed**: 4 of 6 baseline seeds also learned the shortcut (label-following above 50%). In RoBERTa, default training with fixed "true"/"false" labels is enough to produce it in most seeds. The [MASK]→label block brings 11 of 12 runs to 12% or less.

RoBERTa-large (lr 1e-5)	BoolQ accuracy	Label-following (bench 1)	With [MASK]→label block	Accuracy, none → question→passage blocked
planted, seed 0	0.790	97.8%	6.0%	0.602 → 0.547
planted, seed 1	0.817	82.3%	2.8%	0.735 → 0.597
planted, seed 2	0.820	92.4%	37.6%	0.743 → 0.587
planted, seed 3	0.747	89.6%	10.5%	0.658 → 0.567
planted, seed 4	0.710	56.5%	1.2%	0.632 → 0.543
planted, seed 5	0.713	47.0%	3.0%	0.563 → 0.553
baseline, seed 0	0.803	66.8%	10.2%	0.688 → 0.592
baseline, seed 1	0.773	4.6%	5.4%	0.610 → 0.543
baseline, seed 2	0.813	6.7%	2.3%	0.770 → 0.572
baseline, seed 3	0.797	71.7%	3.6%	0.695 → 0.545
baseline, seed 4	0.780	66.9%	7.6%	0.725 → 0.567
baseline, seed 5	0.783	76.7%	11.8%	0.758 → 0.582

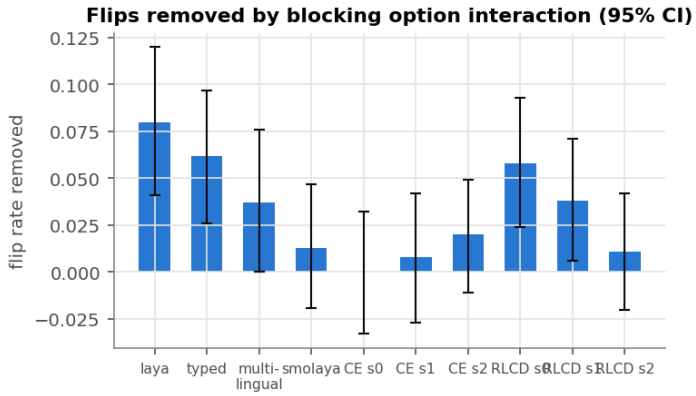
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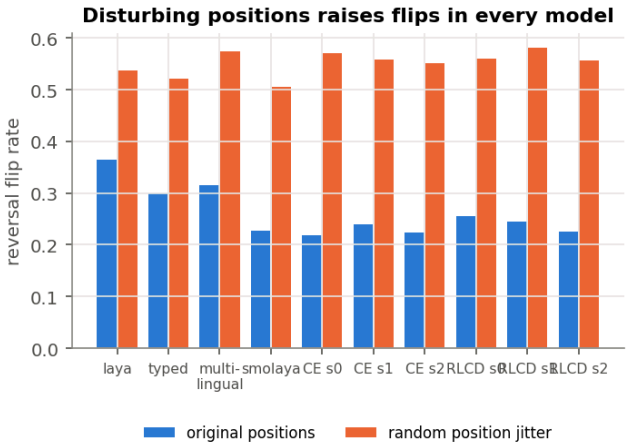
5 · Supporting findings

5.1 C1: order sensitivity is positional; competition is separate

Reversing the options flips 29.6% of laya's multiple-choice answers (fp32, 1,392 items), and options compete: blanking one shifts another's centred logit by 0.37. A factorial that holds [MASK]–option binding fixed shows that blocking option interaction removes only a few points of flips in every model, while order-invariant layouts keep most of the competition.

Condition (laya, fp32)	Accuracy	Reversal flips	Mean abs(M)
Trained layout	0.453	29.6%	0.374
Options cannot see each other	0.414	28.3%	0.181
Options blocked + shared positions	0.408	0.0%	0.175
Fully masked parallel	0.346	28.6%	0.000





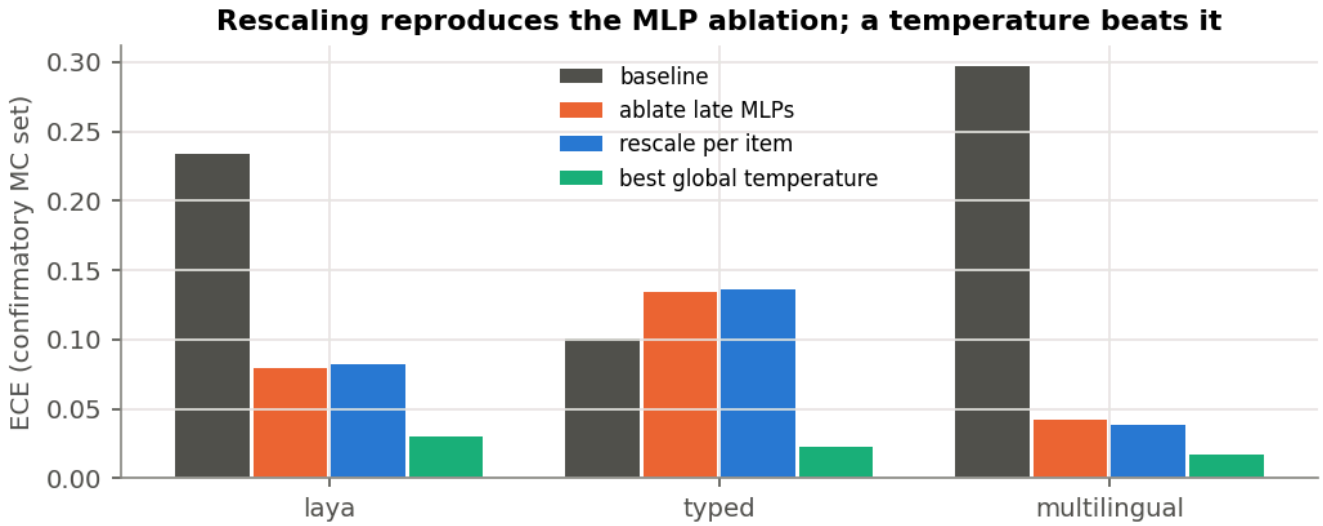
Model	Flips removed by blocking interaction (pts, 95% CI)	Competition kept when order-invariant	Accuracy cost (pts)
laya	8.0 [4.1, 12.0]	78%	1.9
typed	6.2 [2.6, 9.7]	76%	1.4
multilingual	3.7 [0.0, 7.6]	76%	-0.7
smolaya	1.3 [-1.9, 4.7]	77%	4.2
CE s0	0.0 [-3.3, 3.2]	86%	9.7
CE s1	0.8 [-2.7, 4.2]	87%	9.6
CE s2	2.0 [-1.1, 4.9]	90%	8.9
RLCD s0	5.8 [2.4, 9.3]	77%	8.2
RLCD s1	3.8 [0.6, 7.1]	82%	8.6
RLCD s2	1.1 [-2.0, 4.2]	87%	10.4

Giving options reversed position ids without moving tokens reproduces the reversed run in 100% of flips in all 10 models; this holds by construction for a bidirectional encoder and is an identity check, not evidence. The competition measure is robust to the replacement used (filler 0.39, dash 0.40, distractor 0.44).

5.2 C2: confidence is set by a late-MLP amplifier

Removing MLPs 22, 27, 26 halves laya's logit scale (ratio 0.47) and fixes most miscalibration, but rescaling each item reproduces it and one temperature beats it, so by the pre-registered rule these MLPs are an amplifier, not a confidence regulator. With the temperature fitted on the exploratory split (T = 2.61), confirmatory ECE is 0.055 vs 0.080 for MLP ablation.

Laya, confirmatory MC set	Accuracy	ECE	Brier	Answers kept
Baseline	0.364	0.234	0.816	100.0%
Ablate late MLPs	0.372	0.080	0.742	95.8%
Rescale each item to the ablated size	0.364	0.082	0.739	100.0%
Best global temperature	0.364	0.030	0.719	100.0%



5.3 C4: the jailbreak detector reacts to style

- Laya detects natural jailbreaks with AUROC 0.947. A linear probe reads the attack from layer 0, but patching shows the decision is made late, relayed by the question tokens: the same early-decodable, late-decisive timing Wen et al. (2026) report for prompt-injection compliance in decoder LLMs.
- **Style test.** The same 150 harmless requests are flagged 12.7% written plainly and 54.0% written in jailbreak style (persona, drama, no override); for the injection question 2.7% vs 42.7%.
- **Override removal (strict, pre-registered, amended).** gpt-5.6-terra was told to delete only override spans from 145 flagged jailbreaks. The pre-registered deletion-only check passed just 2 edits, because the model echoed our input's "Prompt:" prefix; with the amendment logged before re-scoring (strip the echo, compare words only), 104 are deletion-only and 62 of those were judged override-free by gpt-5.6-luna. Laya still flags 62 of 62 (100%, Wilson 95% CI 94.2%–100.0%); the mean log-odds drops by only 0.81. On manual inspection 3 of these edits keep part of a jailbreak trick; without them laya flags 59 of 59 (CI 93.9%–100.0%). Removing the override barely moves laya. An earlier, looser version of this test (82%) is withdrawn: its edits removed far more than the override.
- Adding the jailbreak direction at the layer-20 [MASK]s raises benign prompts flagged from 23% to 61%.

Unlike the intent-tracking late layer described by Wen et al., laya's decision follows surface style.

5.4 C5: training objective

3 seeds per arm	Cross-entropy	RLCD (laya's loss)	Welch p
Competition, mean abs(M)	0.446 ± 0.041	0.374 ± 0.036	0.09
Calibration error	0.129 ± 0.015	0.103 ± 0.022	0.18
Accuracy	0.709 ± 0.010	0.695 ± 0.007	0.12

RLCD trends toward less competition and better calibration; neither difference is significant with 3 seeds.

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6 · Discussion

For practitioners. When answer options carry words with meaning, an option-scoring encoder can learn to answer from those words. In our retraining this happened whenever the label word identified the option during training, even when the option text made it redundant. Two remedies work in different settings:

- For a deployed laya checkpoint, the attention guard removes the behaviour at a cost of 1.8 accuracy points on benchmark 1 and is available as a drop-in wrapper.
- For a new model, randomising label words during training prevented the shortcut in every seed we trained.

On mechanistic fixes. The guard was derived from laya's internals and transfers to its sibling checkpoints, but not cleanly to every retrained model: the same edge classes carry the label, in different layers, and in some seeds they are entangled with normal answering. A fix found in one model is a hypothesis about the next one, not a guarantee.

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7 · Related work

Order and option-ID bias are documented for decoder LLMs (Pezeshkpour & Hruschka 2024; Zheng et al. 2024), and Anchored Answers traces GPT-2's preference for "A" to MLP value vectors. Set-Based Prompting removes order dependence by sharing positions and masking attention. Label Words are Anchors studies in-context demonstration labels in decoders, a different setting from answer-option labels. Confidence Regulation Neurons describes entropy neurons that are explicitly not a temperature; laya's late MLPs are. Wen et al. (2026) report early-decodable, late-decisive prompt-injection compliance in five decoder LLMs with a late representation that tracks intent; laya shares the timing but not the robustness to style. Seeds learning different circuits is known (A Toy Model of Universality). A BERT cross-encoder BM25 circuit is the closest encoder precedent we found. Notes and the search log are in `LITERATURE_NOTES.md`.

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8 · Limitations

- Laya's training data are unknown; the retraining shows a sufficient cause, not the actual one.
- Laya was not built for multiple-choice QA, so C1 and C2 describe how it handles options, not tuned QA behaviour.
- GPT-made data (contradicting edits, some items, style pairs, override edits) were checked by a second GPT model and by a Claude audit of random samples (Appendix A). The flagged rows have not yet been reviewed by a person.
- Some arms have 3 seeds (D0, D2, D3, the earlier regimes); the training-time fix rests on 3 seeds.
- The direction ablation (4.5) was tested on three models and failed its criterion in all three.
- The strict override test (5.3) needed an amendment to its filter after the first run; both counts are reported.

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9 · Corrections made during the study

- The early-timing account of label reading was wrong; the read is redundant and mostly late.
- "A confidence regulator" became "a logit amplifier" under the pre-registered rule.
- A single-seed "20% less competition" became a non-significant trend with 3 seeds.
- "Label-following comes from training data" became "this kind of training can create it".
- "Same behaviour, different mechanisms" became "the same edge classes in different layers, partly entangled with answering".
- The prediction that a redundant label would not be used was wrong (4 of 6 seeds use it).
- The prediction that an all-layer guard would transfer to 5 of 6 retrained seeds failed (3 of 6).
- "Laya flags 82% after only the override is removed" was withdrawn after an audit; replaced by the strict test in 5.3.
- "The training-origin result replicates in RoBERTa" (from 3 seeds) was wrong: with 6 seeds most RoBERTa baselines also learn the shortcut.
- The planned position-vs-token test was an identity by construction, not evidence.

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10 · Reproducibility and cost

Environment: laya 0.4.1, transformers 5.19.0, torch 2.14.1, datasets 5.1.0, Modal client 1.6.1, Python 3.12, H100 GPUs. Every file under `release/results` is produced by `modal run modal_interp.py --task <name>`, and this paper is rebuilt from those files by `python release/scripts/build_paper.py`. Pre-registrations and outcomes are logged in `NEXT_STEPS_PLAN.md` with their git commits. Modal compute at build time: 66.80 metered, $0.00 billed. OpenAI usage (gpt-5.6-terra, gpt-5.6-luna) for data generation and checking is listed in `release/README.md`.

Appendix A · Audit of model-made data

Sheet (random sample)	Check	Pass	Rate (95% Wilson CI)
Contradicting edits, benchmark 1	edit flips the answer	56/60	93% (84%–97%)
Contradicting edits, benchmark 2	edit flips the answer	58/60	97% (89%–99%)
Contradicting edits, BoolQ	edit flips the answer	57/60	95% (86%–98%)
GPT-written C3 items	gold answer correct	60/60	100% (94%–100%)
GPT-written C3 items	answer determined by passage	57/60	95% (86%–98%)
Style pairs	same request	141/150	94% (89%–97%)
Style pairs	no override in styled version	150/150	100% (98%–100%)
Jailbreak rewrites (first version)	override removed	53/60	88% (78%–94%)
Jailbreak rewrites (first version)	only the override removed	21/60	35% (24%–48%)

Judged by Claude (a model independent of the GPT generators), not by a person; flagged rows are marked for human review in `audit/claude_filled/`. Only the first-version jailbreak rewrites failed the audit; that result was withdrawn and replaced by the strict test.